Min-Chia Tsai

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PERSONAL PROFILE

Research assistant at the Institute of Molecular Biology, Academia Sinica, with a background in Biomedical Science from National Chung Cheng University. Research experience spans cancer biology and RNA-mediated gene regulation, with particular interests in non-coding RNAs, neuronal gene regulation, and the molecular mechanisms underlying neural development and degeneration. Hands-on training in molecular and cellular biology, imaging-based assays, animal work, and experimental troubleshooting has built a strong foundation for mechanistic research and graduate study.

EDUCATION

Bachelor's Degree in Biomedical Science **Sep 2020 – Sep 2025**
National Chung Cheng University, Taiwan

- Received academic training in biomedical science, including molecular biology, biochemistry, cell biology, and related laboratory practice.
- Conducted undergraduate research on the anticancer effects of wogonin in ovarian cancer through the AMPK–TET2–5hmC axis.

RESEARCH EXPERIENCE

Research Assistant **Sep 2025 – Present**
Jun-An Chen Laboratory, Institute of Molecular Biology, Academia Sinica, Taipei, Taiwan

- Work on the functions of non-coding RNAs in neural development and degeneration.
- Participate in a project investigating how the lncRNA Meg3 contributes to long neuronal gene maturation.
- Perform experiments including mammalian cell culture, co-culture assays, qPCR, Western blotting, drug treatment, and RNAscope combined with immunofluorescence.
- Assist with mouse spinal cord dissection, tissue collection, and behavioral experiments.
- Contribute to experimental troubleshooting, protocol refinement, and research discussion.
- Provide administrative and coordination support for laboratory operations.

Administrative Assistant **Jun 2024 – Jul 2025**
Michael W. Y. Chan Laboratory, National Chung Cheng University, Taiwan

- Supported laboratory administration, organization, and general coordination.
- Assisted with documentation, scheduling, and routine laboratory operations.

Undergraduate Research Student **Sep 2022 – Jun 2024**
Michael W. Y. Chan Laboratory, National Chung Cheng University, Taiwan

- Investigated the anticancer effects of wogonin in ovarian cancer through the AMPK–TET2–5hmC axis.
- Performed cancer cell culture, qPCR, Western blotting, transfection, and drug treatment experiments.
- Participated in molecular and cellular experiments, data collection, and scientific discussion in cancer biology research.

RESEARCH SKILLS

Molecular and Cellular Biology	Mammalian and cancer cell culture; co-culture assays; qPCR; Western blotting; drug treatment assays; DNA/RNA extraction; transfection; gel electrophoresis.
Imaging and Tissue-Based Techniques	RNAscope combined with immunofluorescence; confocal microscopy; mouse spinal cord dissection and tissue collection; tissue processing for downstream analysis.
Animal Work	Mouse behavioral experiments.
Research Workflow	Experimental troubleshooting and protocol optimization; scientific literature reading and figure interpretation; journal club and lab meeting presentations; scientific slide preparation; experimental record keeping and data organization.

PUBLICATIONS

Published. Yang, S.-Y., Jhang, J.-S., Huang, W.-L., Tsai, L. H. J., **Tsai, M.-C.**, Chan, C.-P., Lin, R.-I., Lin, H.-Y., Li, C., Yeh, C.-C., & Chan, M. W. Y. *Wogonin Inhibits Ovarian Cancer by Activating the AMPK-TET2-5hmC Axis. Molecular Carcinogenesis*, 2024. DOI: 10.1002/mc.23856.

Submitted. Chen, K.-W., Fan, H.-C., Huang, G.-L., Su, Y.-C., Luo, J.-D., Wang, T.-H., Yen, Y.-P., Yang, Z.-D., Chen, T.-D., Hsieh, C.-Y., Yang, J.-H., Li, J., Liau, E. S., Huang, Y.-T., Lee, Y.-H., Li, L., Chang, Y.-C., **Tsai, M.-C.**, Tian, X., Chen, B.-C., Hsia, K.-C., Hung, J.-H., & Chen, J.-A. *Meg3 lncRNA Organizes ELAVL Condensates to Facilitate Long Neuronal Gene Maturation.* Manuscript submitted.

POSTER PRESENTATION

Investigation of Meg3-ELAVL2 mediated alternative splicing. Min-Chia Tsai, Yi-Ching Su, Kuan-Wei Chen, Jun-An Chen. Poster presented during a summer internship at the Institute of Molecular Biology, Academia Sinica, Taipei, Taiwan.

UNDERGRADUATE RESEARCH PROPOSAL

Investigating the molecular mechanism of SLC25A22 in the immunosuppression of gastric cancer. Undergraduate research proposal, 2023.

SELECTED TRAINING & CERTIFICATIONS

- **Academic Research Ethics Education Courses**, Center for Taiwan Academic Research Ethics Education, Certificate of Completion, 6 hours, issued May 20, 2022.
- **GCP Training**, National Taiwan University Hospital Clinical Trial Center, including *The Latest Regulatory Amendments in Clinical Research* and *The Potential Conflict of Interest in Clinical Research: Regulation and Management* (2 total hours, Apr 2022).
- **Laboratory Safety and Hazard Communication Training**, Academia Sinica, 6 hours, completed Sep 17, 2025.
- **AS-IACUC Education Training Course**, Academia Sinica: Annual Briefing on Animal Usage Reporting, attended Mar 12, 2026.